



Metamaterial design based on autoencoder representation using an active learning method

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Abstract

This work introduces a novel, gradient-free metamaterial design method based on Gaussian process regression to represent the density field of a unit cell. The dimension of the design space is determined by the covariance matrix dimension in the Gaussian process regression. We propose compressing this matrix using an autoencoder, enabling the decoder to generate the density field and effectively reduce the originally large design space to a lower-dimensional subspace. In this compressed space, we employ an active learning method, Bayesian Adaptive Direct Search (BADS), for efficient exploration of the design space. We demonstrate that for simple 2D designs aimed at maximizing unit cell stiffness, our method yields results comparable to those of standard topology optimization. Furthermore, we extend our approach to various mechanical problems, from linear elasticity to hyperelastic large deformation and elasto-plasticity under finite deformation, to 3D metamaterial design. This illustrates the method's versatility and effectiveness across a range of applications.

Keywords Metamaterial design · Autoencoder · Active learning method · Bayesian optimization

1 Introduction

Topology optimization (Bendsoe and Sigmund 2003) is a powerful tool for designing complex components, often outperforming traditional design methods. Its effectiveness is particularly notable in creating structures with repeated unit cells, also known as periodic microstructures, metamaterials, or lattice structures (Osanov and Guest 2016). This approach is effective in designing materials with unique macroscopic properties, a process often referred to as “inverse homogenization,” originally developed by Sigmund (1994, 1995), this method has been utilized to create microstructures with tailored elastic properties, including those with negative Poisson's ratios (Larsen et al. 1997; Clausen et al. 2015).

The unique aspect of these periodic cellular materials is that both their microstructural geometry and base material

properties influence their overall effective properties. This characteristic makes them highly customizable in design. Additionally, the advent of advanced additive manufacturing techniques has significantly expanded the possibilities for creating architected materials with intricate microstructures, further broadening the range of material properties that can be achieved. Topology optimization goes beyond simple shape design; it optimizes both the shape and the topology of materials and structures to enhance their performance while adhering to specified constraints. Since the pioneering work of Bendsoe and Kikuchi (De Geus et al. 2017), topology optimization has evolved rapidly over the past 30 years. Today, it encompasses several widely used methods: the Solid Isotropic Material with Penalization (SIMP) method (Bendsoe 1989; Rozvany et al. 1992), the Level-Set method (Wang et al. 2003; Allaire et al. 2004), and the Evolutionary Structural Optimization (ESO) method (Xie and Steven 1993; Huang and Xie 2007), each with their variants addressing specific design challenges. Additionally, recent years have seen the proposal of various feature control methods designed to manage geometric complexity and ensure manufacturability (Guo et al. 2014; Norato et al. 2015, 2004).

In our exploration of cellular materials with periodic microstructures, we focus on evaluating their effective

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material properties at the microscale. This evaluation is based on the smallest unit of repetition, known as the base cell, and we compute the effective properties using numerical homogenization methods. A range of topology optimization methods have been developed for the design of these microstructures. Sigmund (1994) conducted a study on the creation of material microstructures, either in truss or continuum forms, with specifically defined positive semi-definite constitutive tensors. Huang et al. (2011) took this further by enhancing the bidirectional evolutionary structural optimization (BESO) method to optimize the design of cellular materials for maximum bulk and shear modulus. In parallel, Takezawa et al. (2017) focused on the design and manufacturing of porous metals aimed at achieving effective isotropic thermal conductivity. The design of composites for extreme thermal expansion and bulk modulus, using multiple base materials, was also explored by researchers like Sigmund and Torquato (1996). These studies have led to the creation of microstructures with intriguing properties, such as negative Poisson's ratios, through topology optimization. Moreover, the design of graded cellular materials and the challenges related to microstructure connectivity have been thoroughly investigated (Zhou and Li 2008; Liu et al. 2020). To aid in educational and research efforts, Xia and Breitkopf (2015) have provided a MATLAB implementation for topology optimization of cellular material microstructures, making this advanced area of study more accessible.

While mathematical programming has significantly advanced topology optimization, effectively addressing a broad spectrum of engineering and applied science applications, the unique strength of gradient-free topology optimization lies in its ability to tackle black-box problems. These are problems where the relationship between inputs and outputs is complex, making traditional gradient-based methods less effective as they rely on known gradients to solve problems. Thus, there is an urgent need for further research in black-box topology optimization. Such research is essential to fully leverage the capabilities of gradient-free methods for solving complex optimization problems, especially in scenarios where obtaining gradients through traditional methods proves challenging. Additionally, the availability of open-source finite element packages like MOOSE (Permann et al. 2020), which facilitate the solving of complex, multi-physics physical problems, highlights the need for advancements. However, automatically obtaining the gradient of the objective with respect to design variables can be difficult for complex simulations of this type. Therefore, it is imperative to develop black-box optimization methods, also known as active learning methods, to effectively explore the design space. Gradient-free topology optimization was introduced by Sandgren and his associates (Clausen et al. 2015; De Geus et al. 2017), applying a binary-coded genetic algorithm (GA) with the traditional 0/1 representation. It is

not surprising that GAs have been applied widely as they are generally less likely to be entrapped in local optima. However, implementation of enhancing mechanisms is required to increase the quality of obtained solutions, improving convergence and reducing computational cost. For instance, Jakiela et al. (Sigmund and Torquato 1996) used fitness sharing for speciation to enhance diversity and added a penalty over the number of holes to generate meaningful structures. Kane et al. (Zhou and Li 2008; Liu et al. 2020) proposed tailored genetic operators for 2D structures. Moreover, the influence of the initial population has been investigated in Permann et al. (2020); Luh et al. 2011). In addition to GAs, other metaheuristics and stochastic approaches have been applied to topology optimization, e.g., Simulated Annealing (Guirguis et al. 2018; Liu et al. 2021; Sandgren and Jensen 1992; Jakiela et al. 2000; Kane et al. 1995), differential evolution (Bureerat and Limtragool 2006), hill-climbing search (Bureerat and Limtragool 2008; Wu and Tseng 2010), and particle swarm optimization (Luh et al. 2011). In recent years, Bayesian optimization approaches have become popular for topology design problems due to their ability to balance exploration of the design space with optimization of a machine learning (ML) surrogate model of the previously collected input-response pairs. Guirguis et al. (2018) demonstrate how the derivative-free level-set method can eliminate this drawback without scarifying the quality of obtained topologies. Kang and Luo et al. proposed material-field series-expansion combined with Kriging surrogate model method to optimize cellular structure and general purpose topology optimization problem (Liu et al. 2021).

In recent years, deep generative models such as Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) have been widely employed in metamaterial design tasks due to their ability to capture complex geometric features and enable inverse design. For instance, Wang et al. (2022) used a Gaussian mixture β -VAE for the design of phononic bandgap materials; Lee et al. (2023) proposed a task-aware latent sampling method to construct diverse metamaterial datasets; and Tian et al. (2024) applied high-dimensional Bayesian optimization on deep latent spaces for inverse design. In parallel, GANs have been used to directly generate microstructure geometries with desired physical properties. Conditional VAEs (cVAEs) have been widely adopted for generating unit cell geometries that satisfy specific physical targets, such as frequency-selective surfaces (Boudagh 2025) and auxetic structures (Zhang and Zhao 2025). VAEs have also been applied to tailor mechanical responses, including custom stress-strain behaviors in irregular metamaterials (Zhang, et al. xxxx), as well as for the vibration control of phononic crystals (Zhao et al. 2025). Beyond simple geometric mapping, recent work incorporates hybrid and ensemble learning frameworks (Xiang et al. 2025), dual VAE architectures (Ko et al. 2025), and

VAE-guided diffusion models (Lyu and Ren 2025), extending the design capability toward more complex and non-parametric metamaterials. Nguyen et al. (2025) introduced property-variational autoencoders (pVAEs) for porous materials, enabling direct mapping between material properties and structure.

Unlike these approaches, our method does not rely on a probabilistic generative model or adversarial training. Instead, we employ a deterministic autoencoder trained on the structured space defined by the inverse covariance matrix of a Gaussian Process (Ω), providing a geometry-aware and spatially smooth low-dimensional latent representation that can be directly optimized using Bayesian methods. Moreover, the proposed pipeline is entirely gradient-free and compatible with black-box forward solvers, making it particularly suited for nonlinear or multi-physics problems where gradient information is difficult or expensive to obtain.

To effectively integrate active learning methods with topology optimization, a key challenge is representing the high-dimensional design space in a lower-dimensional manifold. This is crucial as most active learning methods, like Bayesian optimization, have inherent limitations with high dimensionality. Thus, effectively compressing the design space is a core issue. In recent years, advancements in artificial intelligence algorithms have made significant strides, particularly in data compression techniques. One of the most popular methods in this domain is the autoencoder. Autoencoders are machine learning models designed to learn representation functions for datasets, essentially creating compressed models of the data they learn from. The conventional autoencoder, introduced by Rumelhart et al. (1986), and further detailed by Bourlard and Kamp (1988), comprises two main components: the encoder and the decoder. The encoder's role is to compress and encode the data, effectively transforming it into a reduced latent space. This process, known as the encoding phase, has high value for feature extraction, where it uses the most meaningful information to represent the data. On the other hand, the decoder aims to reconstruct the data from the latent space back to the original representation space, minimizing the error between the original and decoded data. Autoencoders offer enhanced capabilities for feature selection and extraction, especially in deep learning contexts. They can automatically extract target features, addressing the limitations of traditional manual feature extraction methods and helping to avoid overfitting.

In this paper, we propose using an autoencoder method to effectively represent the design space. By applying this approach, we can compress the design space into a low-dimensional space, making it suitable for combining with active learning methods for exploring design possibilities. The remainder of this paper is structured as follows: Sect. 2 outlines the mathematical formulation of the density field

representation based on the autoencoder. Section 3 demonstrates the active learning method and the formulation of the optimization problem. Section 4 discusses several numerical examples, ranging from 2 to 3D problems and from linear to nonlinear mechanics problem. Finally, conclusions are drawn in Sect. 5.

2 Density field representation based on an autoencoder

2.1 Density field representation based on Gaussian process regression

The Fig. 1 illustrates a 1D Gaussian Process (GP) regression over the domain $[0,1]$. The solid black line depicts the GP's expected mean response, while the shaded blue region represents the model's uncertainty. Observed data points, generated by introducing noise to a sine function, are shown as red markers. These observations guide the GP model, with areas further from data points showcasing increased uncertainty. Note that Illustration of the smoothness imposed by a stationary RBF kernel used in the Gaussian Process prior. The shaded band does not represent a conventional posterior confidence interval but rather reflects typical sample variation from the prior.

A Gaussian process is defined by a mean function $m(x)$ and a covariance function (kernel) $k(x, x')$. For simplicity, we often assume a zero-mean function:

$$m(x) = 0 \quad (1)$$

The covariance or kernel function represents the relationship between the function values at two points. The Gaussian (Radial Basis Function or RBF) kernel is one of the most common choices:

$$k(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2l^2}\right). \quad (2)$$

Here, l is the length scale, determining how quickly the correlations decay with distance. Given spatial data points x , the kernel function is applied pairwise to compute the covariance matrix K :

$$K_{ij} = k(x_i, x_j) \quad (3)$$

This matrix represents the relationship between every pair of points in the dataset. For a new data point x^* , the prediction is a Gaussian distribution with mean

$$\mu_* = k_*^T K^{-1} y \quad (4)$$

and variance

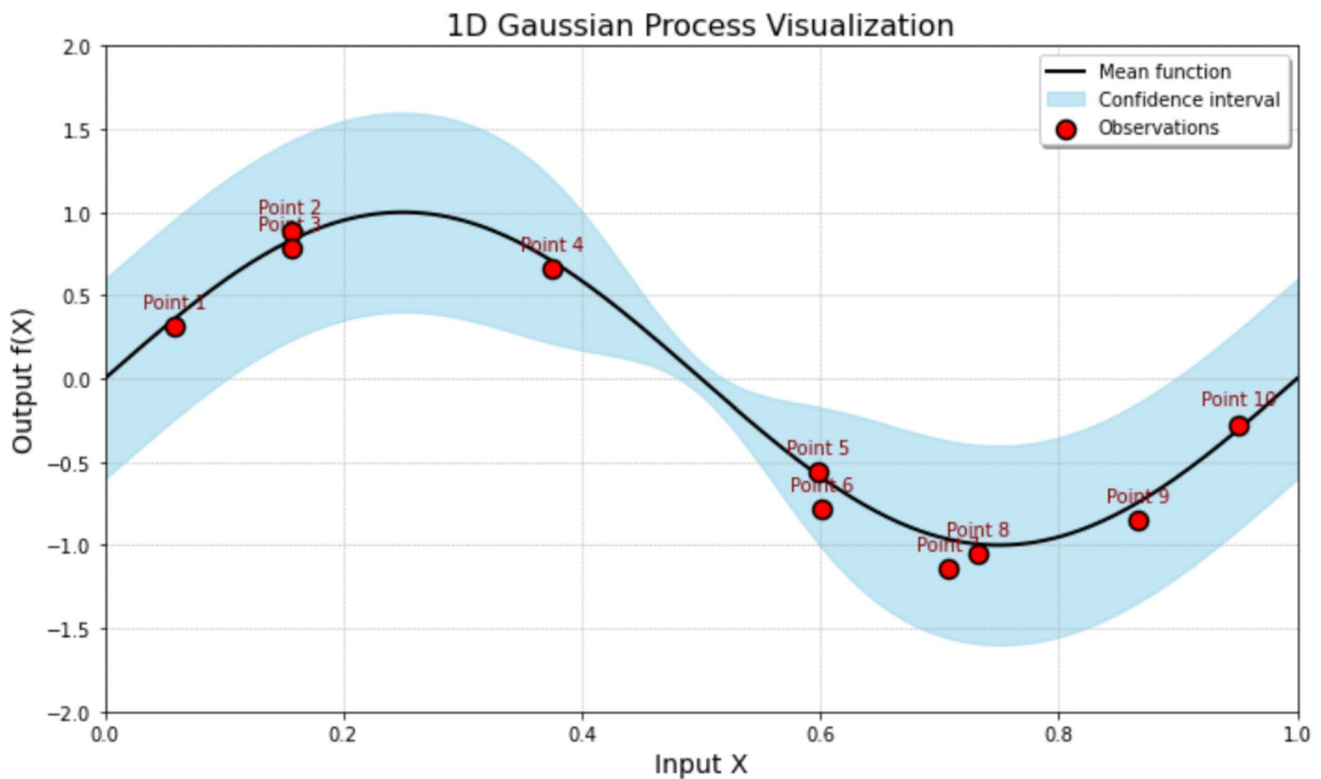


Fig. 1 Gaussian process regression

$$\sigma_*^2 = k(\mathbf{x}_*, \mathbf{x}_*) - \mathbf{k}_*^T \mathbf{K}^{-1} \mathbf{k}_*, \quad (5)$$

where, $\mathbf{k}_*$, the covariance vector between the new point and the observed data points, is given by

$$\mathbf{k}_* = [k(\mathbf{x}_*, \mathbf{x}_1), \dots, k(\mathbf{x}_*, \mathbf{x}_n)]^T, \quad (6)$$

where $\mathbf{x}_*$ represents the density value at the inquiry point. Assume we have a 3D density field ρ . We know several density values at specific spatial coordinates $\mathbf{x}_i (i = 1, 2 \dots N)$, where each $\mathbf{x}_i$ can represent the centroid of finite elements in the FE method. Consequently, our observations, denoted as ρ_c , correspond to the density values at these points. We can disregard the variance since we view the density field as deterministic. Relying on the Gaussian process described earlier, we can reconstruct the complete density field as follows:

$$\rho = \mathbf{k}_*^T \mathbf{K}^{-1} \rho_c. \quad (7)$$

We can treat the observed ρ_c as design variables meant for optimization. Here, we introduce a vector λ , where

$$\mathbf{K}^{-1} \rho_c = \lambda \quad (8)$$

Assuming that $\mathbf{\Omega} = \mathbf{K}^{-1}$, the equations above can be further rewritten as

$$\sum_{i=1}^N \mathbf{\Omega}[:, i] \cdot \rho_c[i] = \lambda. \quad (9)$$

Here, $\mathbf{\Omega}[:, i]$ represents the i th column vector of the matrix $\mathbf{\Omega}$, and i is the i th value of the vector ρ_c . Now, the vector λ becomes a linear combination of the column vectors of the matrix $\mathbf{\Omega}$. In the subsequent section, we elucidate how an autoencoder can be employed to learn the matrix $\mathbf{\Omega}$, and the vector λ can be directly generated by the post-trained decoder. Using functions to describe the density field has been extensively researched, particularly in the context of the level-set method. Here, Radial Basis Functions (RBF), for instance, are commonly employed to represent the level-set function (Guirguis and Aly 2016b; Wang and Wang 2006). However, the integration of the Gaussian process with density field representation is a novel proposition. Unlike recent works such as Yousefpour et al. (2025), which place GP priors on both design and state fields and embed them within a physics-informed optimization loop, we do not use GP as a differentiable surrogate model. Instead, the inverse covariance matrix $\mathbf{\Omega}$ is used once to define a spatially smooth latent design space, which is then compressed and explored via a gradient-free method. To our knowledge, this reinterpretation of GP structure for latent design representation has not been reported before (Fig. 1).

2.2 Autoencoder with multiple layers

An autoencoder is a specialized artificial neural network designed for unsupervised learning of efficient data encodings. It learns two primary functions: an encoder that compresses the input data, and a decoder that reconstructs the original data from this compressed representation. The primary objective of an autoencoder is to learn a compact representation of the dataset, often for the purpose of dimensionality reduction. A schematic diagram illustrating the structure of an autoencoder can be seen in Fig. 2.

2.2.1 Encoder

Given an input vector x , the encoder produces a latent (compressed) representation z . For each layer l in the encoder, the transformation is

$$h_l^e = \varphi(W_l^e h_{l-1}^e + b_l^e), \quad (10)$$

where $h_0^e = x$ is the input to the encoder. W_l^e and b_l^e are the weight matrix and bias vector for the l th layer of the encoder, respectively. φ is the activation function, e.g., sigmoid, tanh, ReLU, etc. h_l^e is the output of the l th layer. The latent representation z is given by

$$z = h_L^e \quad (11)$$

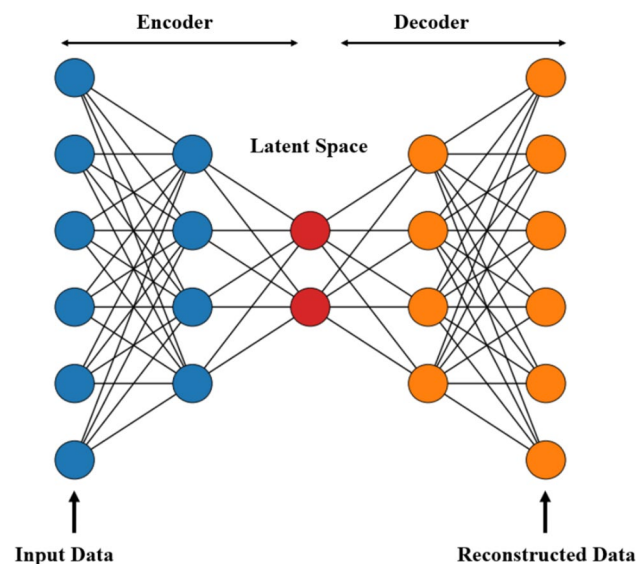


Fig. 2 The structure of an Autoencoder

2.2.2 Decoder

The decoder reconstructs the input from the latent representation z . For each layer l in the decoder, the transformation is

$$h_l^d = \psi(W_l^d h_{l-1}^d + b_l^d), \quad (12)$$

where $h_0^d = z$ is the input to the decoder (i.e., the output of the encoder). W_l^d and b_l^d are the weight matrix and bias vector for the l th layer of the decoder, respectively. ψ is the activation function for the decoder, which could be the same as φ or different. h_l^d is the output of the l th layer. The final reconstruction x_{hat} is

$$x_{hat} = h_L^d. \quad (13)$$

The objective remains to minimize the difference between the original input x and its reconstruction x_{hat} . The Mean-Squared Error (MSE) is a common choice:

$$L(x, x_{hat}) = \frac{1}{n} \sum_{i=1}^n (x_i - x_{hat_i})^2 \quad (14)$$

It is important to clarify that during training, the autoencoder is not directly learning to generate the vector λ used in optimization. Instead, the decoder is trained to reconstruct individual column vectors from the inverse covariance matrix Ω , with each training sample corresponding to a single column. The loss function (Eq. 14) is computed between each original column and its reconstruction. After training is complete, the decoder is repurposed as a nonlinear generator that maps latent vectors z to output vectors λ , which are then used in the density field reconstruction defined in Eq. (16).

To train the described autoencoder, we employ the Adam (Kingma and Ba 1412) optimization algorithm. The Adam solver combines the advantages of two other widely used optimization methods: AdaGrad and RMSProp (Mukkamala and Hein 2017). The primary goal during training is to minimize the reconstruction loss, which measures the difference between the original inputs and their reconstructions produced by the autoencoder. Leveraging gradient descent and back-propagation to iteratively adjust the weights and biases in both the encoder and decoder, the Adam solver dynamically updates the learning rates for each parameter.

We aim to utilize an autoencoder for the representation of the matrix $\Omega_{n \times n}$. For our approach, we treat each column of the matrix $\Omega_{n \times n}$ as a “snapshot.” Given that there are n columns, we have n distinct snapshots. Each of these snapshots serves as an individual input for the autoencoder, leading to a total of n training samples. The primary goal of using the Autoencoder is to compress the matrix $\Omega_{n \times n}$ into a smaller latent space. Once trained based on Eq. (14), we can leverage the latent space variables to reconstruct the column space of matrix $\Omega_{n \times n}$. A critical

point to note is that λ is essentially a linear combination of the columns $\Omega_{n,i} (i = 1, \dots, n)$ of the matrix $\Omega_{n \times n}$. This implies that λ exists within the space spanned by the column vectors of matrix $\Omega_{n \times n}$. Once trained, the decoder acts as a nonlinear generator that maps latent vectors $\mathbf{z}$ to output vectors $\lambda \in \mathbb{R}^n$, each having the same dimensionality as a column of Ω . During training, each column of Ω serves as an individual training sample, and the autoencoder minimizes the reconstruction loss between the original and reconstructed columns. After training, the decoder is repurposed to generate new vectors λ from arbitrary latent inputs $\mathbf{z}$, enabling structured design representations for optimization. The mapping from the latent space to the output vector can be expressed as follows:

$$\lambda = \psi \left(\mathbf{W}_L^d \mathbf{h}_{\{L-1\}}^d + \mathbf{b}_L^d \right), \quad (15)$$

where $\mathbf{W}_L^d$ and $\mathbf{b}_L^d$ denote the weight matrix and bias of the final decoder layer, respectively, and $\mathbf{h}_{L-1}^d$ is the output from the previous layer. This formulation describes the final step of the decoder used to reconstruct or generate individual column-like vectors λ that resemble the structured basis within Ω .

In other words, starting from the latent representation $\mathbf{z}$ (which is $(\mathbf{h}_0^d)$), the vector λ is obtained by iteratively applying the transformations of each layer in the decoder, culminating in the output of the last layer $\mathbf{h}_L^d$. In the above equations, $\mathbf{W}_L^d$ is the weight matrix associated with the last layer, L , of the decoder. It dictates the strength and direction of the connection between the neurons of the $(L-1)$ th layer and the L th layer. $\mathbf{h}_{\{L-1\}}^d$ represents the output from the $L-1$ th layer of the decoder. Essentially, it is the transformed version of the latent representation after being processed by the preceding layers of the decoder. $\mathbf{b}_L^d$ is the bias vector for the last layer, L , of the decoder. The bias allows the model to adjust its output independently of its input, providing an added degree of freedom. Thus, we can reconstruct the density ρ (shown in Eq. (7)) based on decoder representation as follows:

$$\rho_r = \mathbf{k}_*^T \lambda = \mathbf{k}_*^T \psi \left(\mathbf{W}_L^d \mathbf{h}_{\{L-1\}}^d + \mathbf{b}_L^d \right) \quad (16)$$

In this section, an autoencoder efficiently compresses the matrix $\Omega_{n \times n}$ by treating each of its columns as distinct snapshots. The nonlinear nature of the decoder enables the direct generation of the vector λ from the latent space. This suggests that any linear transformation, such as λ , inherently resides within the subspace created by the autoencoder's decoder. Leveraging this insight, we successfully reconstructed the density ρ from the decoder's representation.

The matrix $\Omega_{n \times n}$ must be normalized within the range of $[0,1]$ before training with the autoencoder; this step is

imperative for the machine learning framework to function effectively. However, the reconstructed density field, as given by Eq. (16), may sometimes fall outside the $[0,1]$ range. To address this, we apply a mathematical transformation to ensure the output density ρ_p is appropriately constrained. The transformation is defined as follows:

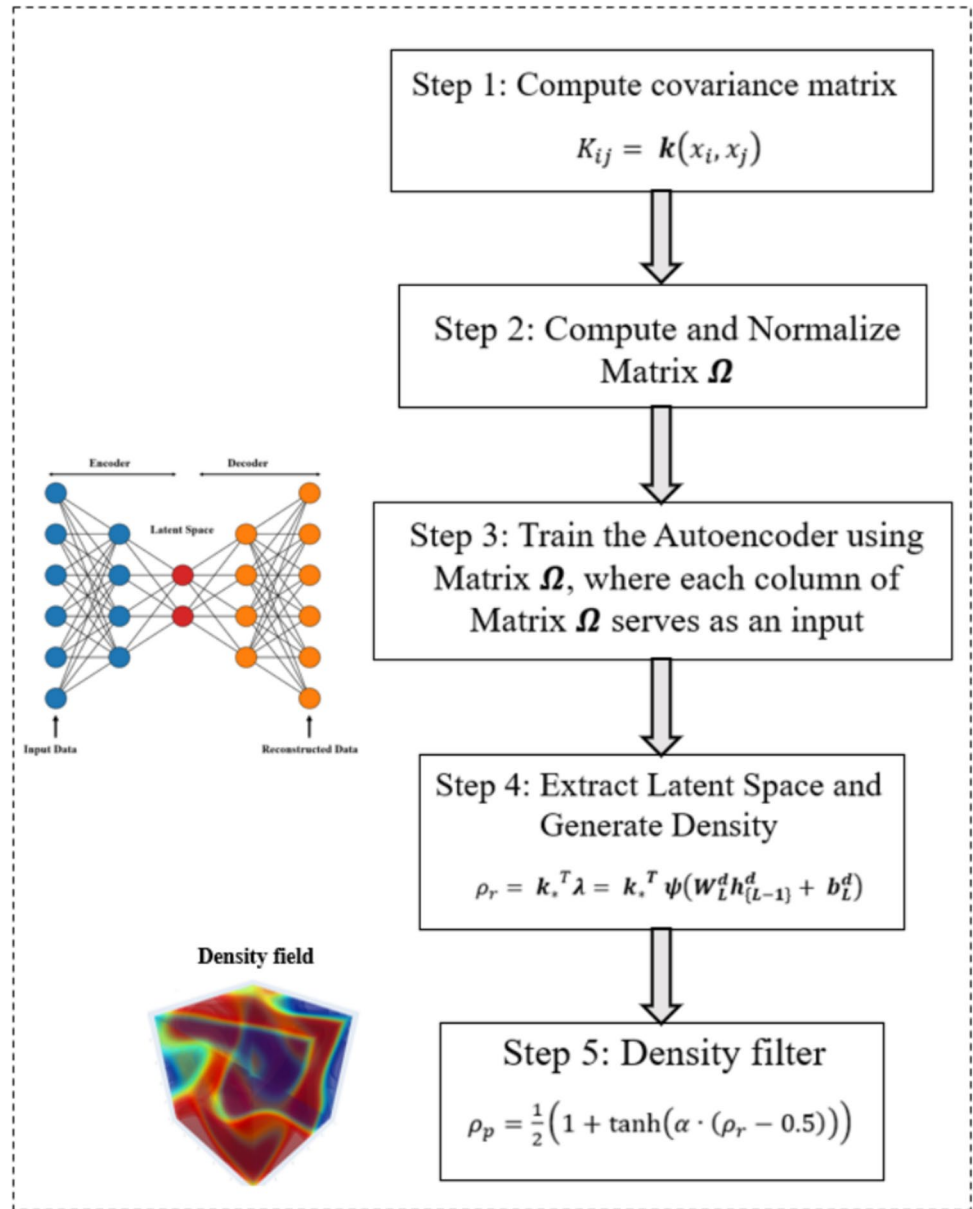
$$\rho_p = \frac{1}{2} \left(1 + \tanh(\alpha \cdot (\rho_r - 0.5)) \right). \quad (17)$$

Here, α is an adjustable parameter that controls the steepness of the transformation. For the purposes of this paper, we have selected $\alpha = 15$ to effectively confine the output density ρ_p within the desired range (Fig. 3).

As demonstrated in the figure above, the transformation ensures that the effective density for FEM analysis falls within the range of $[0,1]$. In practical applications, the minimum value of ρ_p is set to a small threshold, such as 1×10^{-3} , to prevent computational singularities. The flowchart for generating density based on the Autoencoder can be described as follows: First, compute the covariance matrix utilizing finite element mesh information. Subsequently, calculate and normalize the matrix Ω . Then, train the Autoencoder using each column (snapshot) of matrix Ω . From this, extract the latent space and utilize it to reconstruct matrix Ω , enabling the reconstruction of the density ρ_r . Finally, filter the density ρ_r to ensure that the output density falls within the range of $[0,1]$. The basic flowchart of this density generation process is depicted in Fig. 4.

It is worth noting that a linear autoencoder with mean-squared reconstruction loss is mathematically equivalent to performing Singular Value Decomposition (SVD) or Principal Component Analysis (PCA) (Fournier and Aloise 2019; Wang et al. 2016). However, for complex geometries, the SVD eigenvalue spectrum decays slowly, requiring many modes to retain sufficient variance, which leads to a high-dimensional design space. In contrast, the nonlinear autoencoder can capture higher-order correlations and achieve stronger compression, yielding a compact latent space that facilitates efficient Bayesian optimization under the curse of dimensionality. A more detailed quantitative comparison between linear and nonlinear compression approaches will be investigated in future work.

As commonly reported in VAE-based microstructure design literature, representing raw voxel-level geometries typically requires a relatively large latent dimension (often 30–60 variables) due to the sharp material–void interfaces and highly non-smooth geometric patterns. In contrast, our method does not encode the raw geometry directly. Instead, the autoencoder operates on the structured and spatially smooth columns of the GP inverse covariance

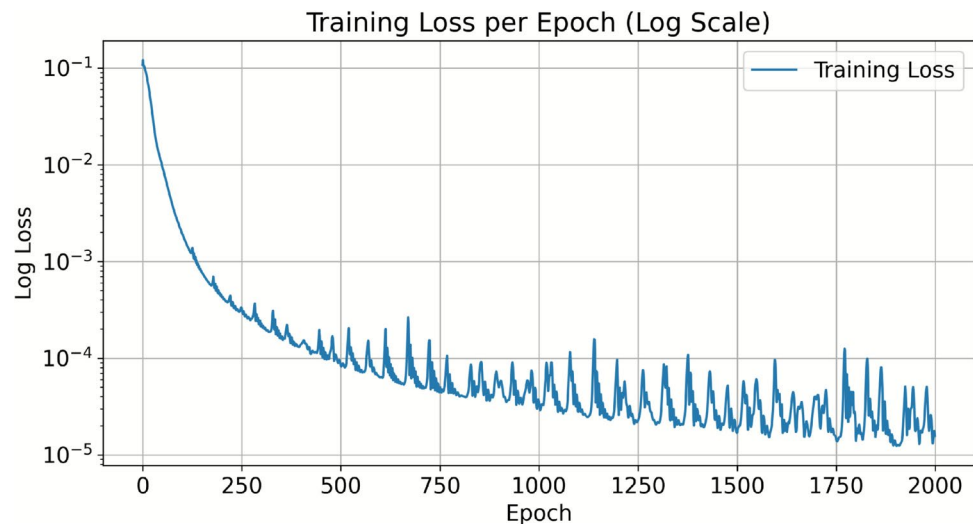
Fig. 3 Flowchart of density generation based on Autoencoder

matrix Ω , which already embeds spatial correlations through the RBF kernel. Because these columns form a smooth functional basis, they are substantially easier to compress. Consequently, our autoencoder achieves accurate reconstruction using only 20 latent dimensions. This highlights a key advantage of incorporating the GP structure: it provides a geometry-aware representation that reduces the intrinsic dimensionality of the design space before nonlinear compression is applied.

3 Optimization formulation and active learning

3.1 Bayesian adaptive direct search

Bayesian optimization is a probabilistic model-based approach for finding the minimum of a function. It builds a probabilistic model of the objective function and uses it

Fig. 4 Training loss of Autoencoder model

to select the most promising points to evaluate. The Bayesian aspect comes from using prior knowledge about the function and updating this with the evidence from function evaluations to get a posterior that guides the search.

The origins of Bayesian optimization can be traced back to the work of Kushner (1964), who adopted Wiener processes for solving unconstrained one-dimensional optimization problems by maximizing the probability of improvement to select subsequent samples. In recent years, various versions and modifications of the standard Bayesian method have been proposed to enhance the performance of Bayesian optimization. Shahriari et al. (2015) provided a comprehensive review of the fundamentals of Bayesian optimization, detailing statistical modeling and popular acquisition functions. Furthermore, Frazier (2018) explored some of the latest advances in Bayesian optimization, particularly in the areas of multi-fidelity and constrained optimization.

The Bayesian Adaptive Direct Search (BADs) algorithm was developed by Acerbi and Ma (2017). This novel Bayesian Optimization (BO) method, along with an associated toolbox, was created with the aim of efficiently fitting moderately expensive computational models “out-of-the-box.” Their paper demonstrates that on real model-fitting problems, BADs outperforms widely used and state-of-the-art methods for non-convex, derivative-free optimization, including ‘vanilla’ BO.

BADs combines Bayesian techniques with a mesh adaptive direct search, initiating with an approximate solution and progressively refining it. The algorithm’s core comprises an iterative search-poll mechanism that leverages Gaussian Process (GP) regression to inform its search strategy. At each iteration, BADs evaluates the objective function, updating its GP model and hyperparameters as necessary. When direct search fails to yield improvements, the algorithm transitions to a polling phase, where a set of candidate

solutions is assessed opportunistically. The adaptive nature of BADs is highlighted by its dynamic mesh and poll sizes, which are doubled following successful iterations to accelerate convergence, or halved when progress stalls, ensuring thorough exploration of the solution space. Termination occurs when either the maximum number of evaluations is reached, the poll size diminishes below a threshold, or no significant advancements are made, culminating in the selection of the most optimal solution encountered. The high-level overview of the BADs algorithm is presented in the form of pseudo-code as follows:

Algorithm 1 BADs Algorithm

Input: objective function f , starting point x_0 , bounds LB, UB
Output: Optimized solution x_{end}
 Initialize mesh and poll sizes, $k \leftarrow 0$
repeat
 Update Gaussian Process (GP) approximation if necessary
 Perform local Bayesian optimization step
 Evaluate f and update the solution if improvement is found
 Adjust mesh and poll sizes based on success of iteration
 $k \leftarrow k + 1$
until termination criteria are met
return Optimized solution x_{end}

The robust performance of BADs is attributable to the interplay between an aggressive search strategy, based on local Bayesian Optimization (BO), and a failsafe poll stage, which guards against failures in the Gaussian Process (GP) surrogate model. More details regarding the implementation of the algorithm can be found in Ref. (Acerbi and Ma 2017).

3.2 Optimization formulation

The general mathematical formulation for using the autoencoder model for metamaterial design can be expressed as follows:

$$\min_{\mathbf{z}} f(\mathbf{u}, \rho(\mathbf{z}))$$

$$s.t. \begin{cases} g_k(\mathbf{u}, \rho(\mathbf{z})) \leq 0, (k = 1, 2, \dots, n_c) \\ \mathbf{G}(\mathbf{u}) = 0 \end{cases} \quad (18)$$

Here, $\mathbf{u}$ is the structural response vector, f and g_k are the objective and constraint functions, respectively, n_c is the number of constraints, and $\mathbf{G}(\mathbf{u}) = 0$ denotes the equilibrium equations of the physical problem. Notably, since we use the BADS method, which can only handle unconstrained problems, we transform the original optimization problem in (18) into an unconstrained problem using a simple penalty method:

$$\min_{\mathbf{z}} f(\mathbf{u}, \rho(\mathbf{z})) + p \cdot \max_k (g_k, 0). \quad (19)$$

In this equation, p is a penalty constant, set at 10^4 for all examples in this paper. The penalty is selected based on several numerical tests; a small penalty leads to violated constraints, while too large a value results in numerical issues. We verified that the optimized designs satisfied the volume constraint with negligible violation. In all cases, the final volume fraction was within 0.5% of the specified limit. The second term implies that any violation of constraints will incur a penalty, making the objective function larger. This ensures that during optimization, the optimizer seeks solutions that satisfy all constraints. After this transformation, we can successfully apply the BADS algorithm. For general purposes, we set all latent space values in the range of $[-5, 5]$, a range carefully selected after numerical testing.

3.3 Geometric symmetry

Geometry symmetry is an essential aspect of our unit cell design algorithm. For 2D problems, implementing symmetry constraints is relatively straightforward. Readers interested in the details of this implementation can refer to Xia and Breitzkopf (2015) for a more comprehensive presentation. In the case of 3D problems, we apply symmetry constraints to ensure the resulting structure tiles space. The fundamental approach to generating a symmetric density field involves the following steps:

- **Initial Density Field Generation:** We begin by using the autoencoder representation to generate a density field Φ for one-eighth of a unit cell.
- **Density Filtering:** This density field Φ is then processed through a density filter, denoted as $\mathcal{F}(\Phi)$
- **Creating the Full Density Field:** The final density field for the entire unit cell is constructed by strategically flipping the density matrix of $\mathcal{F}(\Phi)$. This process mirrors the one-eighth segment to form the complete unit cell density field, ensuring symmetry throughout.

To further demonstrate the density filter F , we assume that the matrix generated by the autoencoder is denoted as $\mathbf{R}$, of size $n \times n \times n$ and $R(x, y, z)$ represents the element of Matrix $\mathbf{R}$ at position (x, y, z) . The symmetric matrix, denoted as $\mathbf{S}$, is formed as follows:

$$S(x, y, z) = 1/6 \cdot (R(x, y, z) + R(y, x, z) + R(z, y, x) + R(x, z, y) + R(y, z, x) + R(z, x, y)) \quad (20)$$

for each x, y, z in the range 1 to n . Here, $S(x, y, z)$ is the average of the values at six symmetrically related points in the matrix R . This equation effectively averages the values of a point and its symmetric counterparts across the three axes and diagonal planes, ensuring symmetry in the resulting matrix S .

This approach ensures that the lattice has translational symmetry. It is somewhat over restrictive for that goal, as the final structure will have orthorhombic symmetry (with equal lattice vectors). Less restrictive symmetry could result in more optimal structures, however, this type of symmetry also ensures the resulting lattices are relatively easy to construct with an additive manufacturing technique.

It is worth noting that VAEs combined with active learning have been widely applied in metamaterial design (Wang et al. 2022; Lee et al. 2023; Tian et al. 2024). Such approaches typically rely on probabilistic sampling, acquisition functions, and iterative retraining to generate or refine candidate designs. By contrast, our framework uses a deterministic autoencoder trained once on the structured basis from the GP inverse covariance, followed by direct optimization in the latent space using BADS. This black-box optimization approach avoids the need for iterative model updating and is particularly advantageous when design evaluations involve expensive nonlinear simulations.

4 Results and discussion

4.1 2D numerical examples

For the two-dimensional example, our goal is to optimize one unit cell to maximize the structural performance. The objective function and corresponding constraint is shown as

$$J = -D_{1212}^H, \quad \text{subject to } v_f < v_f^* \quad (21)$$

where D^H is the homogenized material elastic matrix. Here, our goal is to maximize shear modulus under volume fraction constraint v_f . The analytical formulation of the homogenized stiffness tensor C_{ijkl}^H , which describes the macroscopic behavior of the periodic microstructure, is written as

$$D_{ijkl}^H = \frac{1}{|\Omega|} \int_{\Omega} C_{pqrs} \left(\varepsilon_{pq}^{0(ij)} - \varepsilon_{pq}^{*(ij)} \right) \left(\varepsilon_{rs}^{0(kl)} - \varepsilon_{rs}^{*(kl)} \right) d\Omega, \quad (22)$$

where $\varepsilon_{pq}^{0(ij)}$ is the applied unit macroscopic strain, $\varepsilon_{pq}^{*(ij)}$ and $\varepsilon_{pq}^{*(ij)}$ are fluctuating strain fields in the microstructure, C_{pqrs} is the locally varying elasticity tensor, and $|\Omega|$ is the volume of the unit cell. Further details on the FEM formulation for homogenization are available in the referenced literature (Xia and Breitenkopf 2015). In our numerical implementation, Eq. (22) is solved using the finite element method. Test strains are applied to the boundaries of the base cell following the energy-based homogenization formulation. For more in-depth information, please refer to Ref (Xia and Breitenkopf 2015).

In this case, a two-dimensional finite element mesh is employed, constructed as a structured grid with 100 equally spaced nodes along both the x and y axes, spanning from -0.5 to 0.5 in each direction. This creates a uniform mesh consisting of 10,000 quadrilateral elements. First, utilizing the spatial mesh described above, we calculate the covariance matrix in accordance with Eq. (6) and subsequently derive the corresponding matrix Ω . Upon obtaining matrix Ω , we normalize it within the range [0,1]. Following this normalization, we employ an autoencoder to learn the matrix Ω 's structure and patterns, with the specifics of this process presented in Sect. 2.2.

The autoencoder model comprises an encoder and a decoder. The encoder starts with an input layer connected to a 128-neuron hidden layer using a Gaussian Error Linear Unit (GELU) activation for non-linearity. It compresses data to a 20-neuron latent space, representing the most critical features. The decoder, structured inversely to the encoder, expands from the latent space to a 128-neuron hidden layer with GELU activation, aiming to reconstruct the original input dimensions. The output layer applies GELU again for

normalization. The network is optimized end-to-end to minimize the mean-squared error, balancing information preservation with compression. The Adam optimizer, known for its efficient adaptive learning rate adjustments, is employed for the optimization process. This method dynamically adapts the learning rates based on estimates of lower-order moments of the gradients, which can lead to fluctuations in the learning process. After 2000 iterations, the training concluded with a final training loss of 1.57e-05, as illustrated in Fig. 4. After training, the original matrix Ω was reconstructed from the latent space.

In the process of evaluating the alignment fidelity of the autoencoder-reconstructed column vectors relative to their original counterparts, we employ the cosine similarity metric. This metric quantifies the cosine of the angle between two vectors, thus providing a measure of their directional concordance. Given the high density of similarity values near the apex of unity, indicative of almost perfect alignment, we introduce a metric, the 'Cosine Similarity Loss' ($L_{\cos}$). This metric emphasizes even marginal divergences from the ideal cosine similarity of one. It is defined as the complement of the cosine similarity to one, magnifying subtle deviations and thereby allowing for a more granular analysis of the reconstruction accuracy. The equation for 'Cosine Similarity Loss' is given by

$$L_{\cos} = 1 - \frac{(\mathbf{A} \cdot \mathbf{B})}{(\|\mathbf{A}\| \|\mathbf{B}\|)}, \quad (23)$$

where $\mathbf{A}$ and $\mathbf{B}$ represent the original and reconstructed vectors, respectively. To further elucidate minor discrepancies, we propose to examine this metric on a logarithmic scale, thus rendering minor variances more perceptible and ensuring a rigorous evaluation of the autoencoder's performance. The Cosine Similarity Loss of the trained autoencoder for all

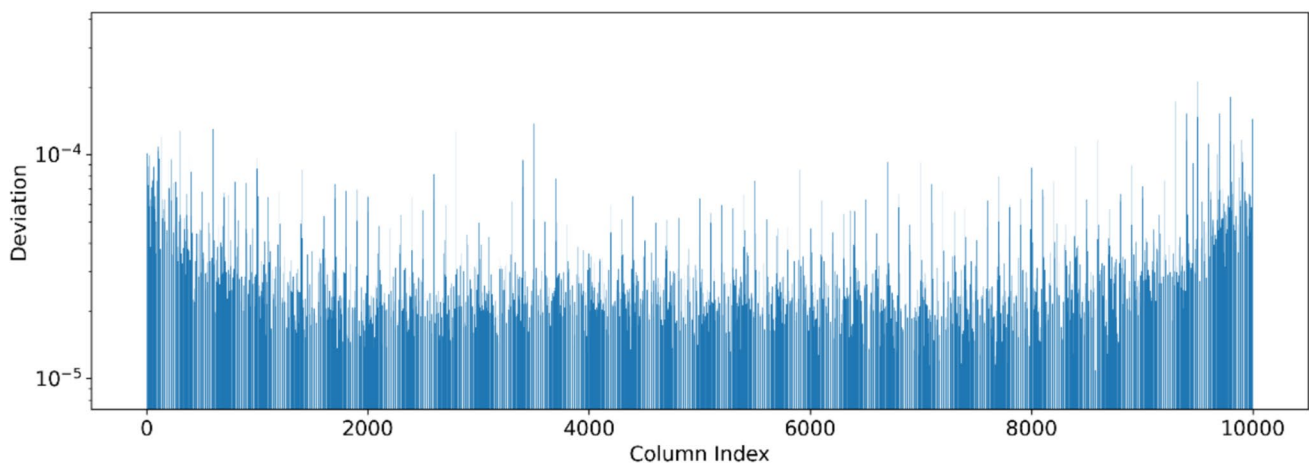


Fig. 5 Cosine similarity loss

Fig. 6 Optimized design with rotational symmetry constraint: **a** unit cell, **b** 5×5 tiling

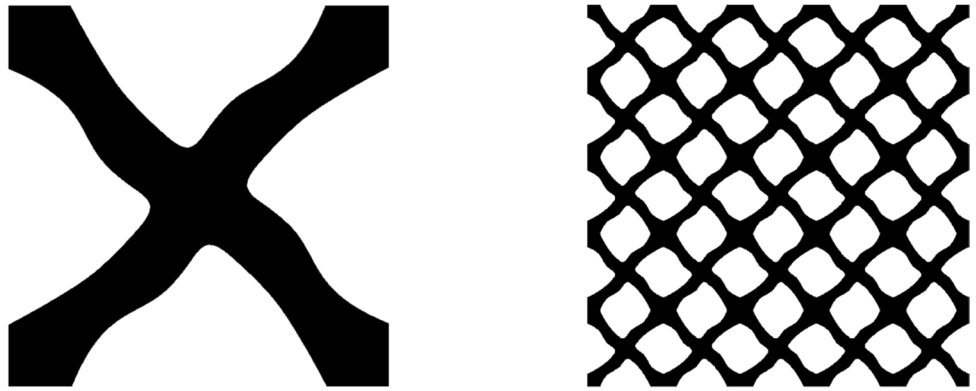
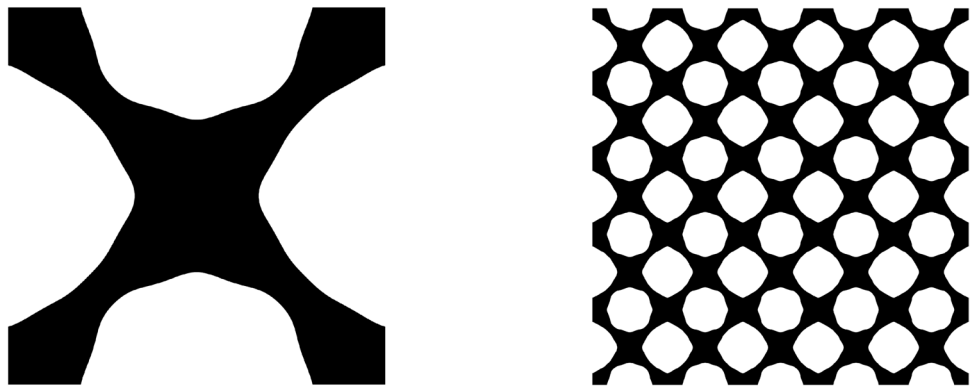


Fig. 7 Optimized design with square symmetry constraint: **a** unit cell, **b** 5×5 tiling



columns is shown in Fig. 5. The maximum deviation value is approximately 10^{-4} , indicating that the columns of the reconstructed matrix align well with those of the original matrix. This suggests that the autoencoder structure captures the inherent characteristics of the matrix. While we do not explicitly separate training and test data, the goal of the autoencoder is not to generalize to unseen samples but to reconstruct the structured basis (columns of Ω) used to generate feasible design candidates. The low reconstruction and cosine similarity losses confirm the decoder's ability to accurately recover the structural features needed for optimization. The effectiveness of this representation is further demonstrated by the design results in Figs. 6 and 7.

The objective of this case study is to maximize the shear stiffness of a unit cell, subject to a volume fraction constraint of less than 0.4, as depicted in Eq. (21). In this instance, rotational symmetry is imposed on the density field. The optimization converged after approximately 771 evaluations, achieving a final optimized objective value of -0.108. The design of a unit cell and an array of 5×5 unit cells are presented in Fig. 6.

Instead of applying rotational symmetry to the density field, square symmetry constraints on the base cell were implemented, with the results showcased in Fig. 7. Compared to the baseline solution from Xia and Breitkopf (2015), (Fig. 8) which uses a hard constraint

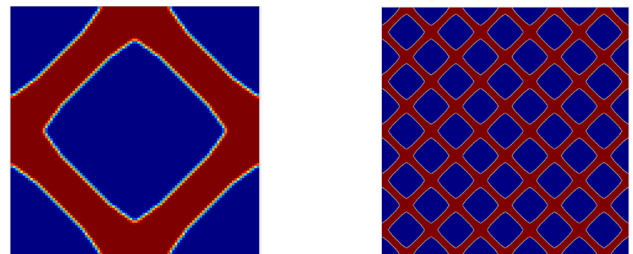


Fig. 8 Optimized design obtained by standard topology optimization method

implementation, our result achieves a nearly identical topology and an objective value within 5% of theirs. This indicates that the penalty method used in our framework is sufficiently effective for enforcing volume constraints (Fig. 8).

We note that the necking effect in Fig. 7, where some bars appear thinner near connections, may be attributed to the limited resolution of the decoder representation or the smoothing behavior of the tanh projection layer. Although the geometry includes these features, the overall mechanical performance remains consistent with the design objectives. In future work, this issue can be alleviated by

increasing the autoencoder latent dimensionality or using a tunable filter radius.

4.2 3D numerical examples

In the 3D scenario, the mesh is structured as a grid with 30 equally spaced nodes in each of the x, y, and z axes, resulting in a total of 27,000 cubic elements. The mesh spans from -0.5 to 0.5 along each axis, forming a uniform cubic element. Based on this 3D spatial mesh, we compute the covariance matrix and subsequently derive the matrix Ω , normalizing it within the range [0,1]. The autoencoder employed in this case is trained to learn and compress the structure of the 3D matrix Ω . The autoencoder architecture for 3D data mirrors its 2D counterpart but scales the layer neurons to 512 to handle the added complexity of 3D inputs. The encoder reduces the data to a 20-neuron latent space, essential for feature extraction, while the decoder reconstructs the 3D matrix from this compact form. Following 3000 training iterations, the autoencoder markedly reduces the mean-squared error, indicative of its effective learning and reconstruction capabilities. The low final training loss underscores its proficiency in accurately capturing and representing the 3D matrix Ω 's intrinsic properties, as illustrated in Fig. 9.

4.2.1 Tension modulus maximization for the linear elastic microstructure design

In this study, we first investigate the linear elastic problem for unit cell design. It is important to note that numerical homogenization is applied to solve general boundary value problems with periodic boundary conditions, as indicated in the reference (Dong et al. 2019). Several papers have worked on this simple unit cell lattice design problem, see references (Mukkamala and Hein 2017; Wang et al. 2019). The homogenized elasticity tensor is denoted by C_{ij}^H . For the asymptotic homogenization method, as described in the references (Dong et al. 2019), where a standard code is provided to perform 3D homogenization for the linear elasticity problem, the implementation of the homogenization code is restricted to analyzing the linear elastic properties of cellular materials. To save computational costs, the unit cell is discretized to 40 by 40 by 40 elements, and the length of the unit cell along the x, y, and z axes is set to 1, respectively. The Lamé parameters, λ and μ , which are the first and second parameters for the solid material from which the lattice structure is made, are set as $\lambda = 0.57$ and $\mu = 0.38$. Periodic boundary conditions are applied. More details regarding the implementation of homogenization are available in the reference material (Dong et al. 2019). The objective in this linear case is to maximize the

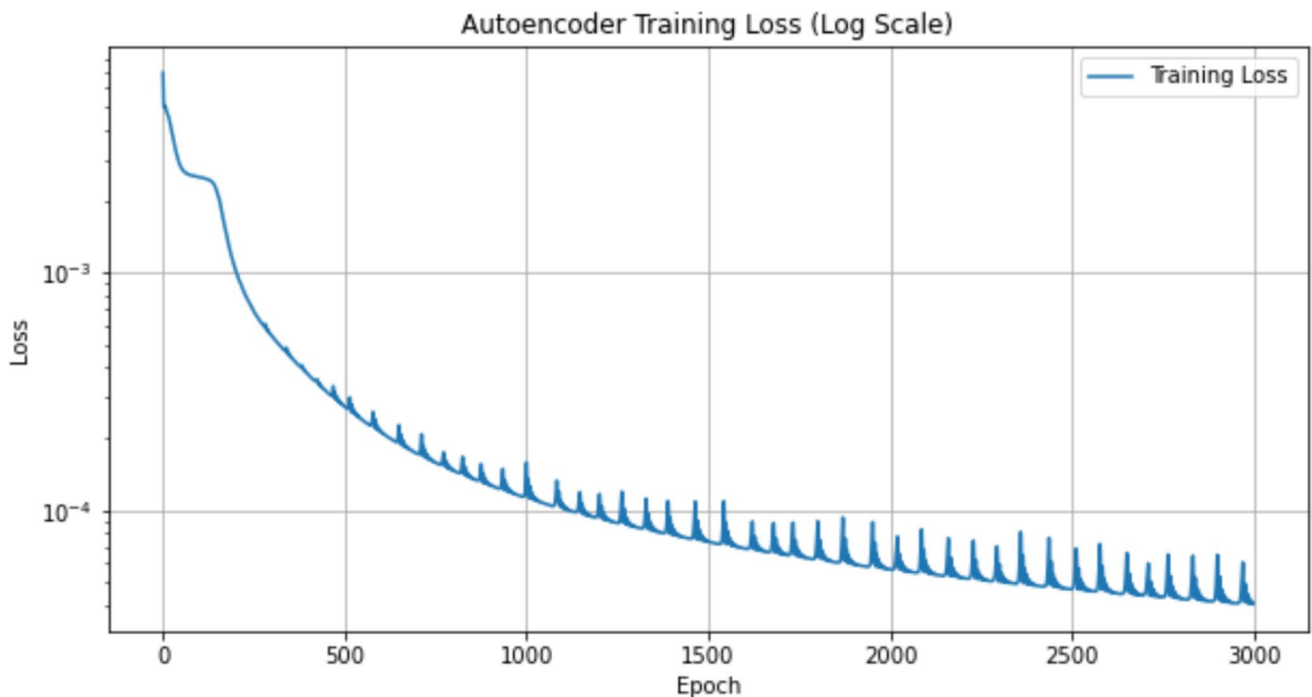
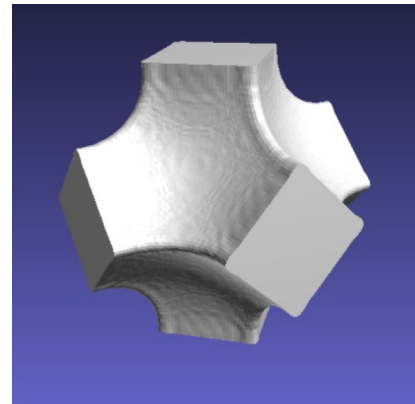
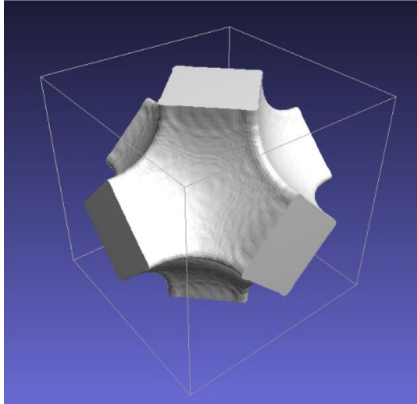
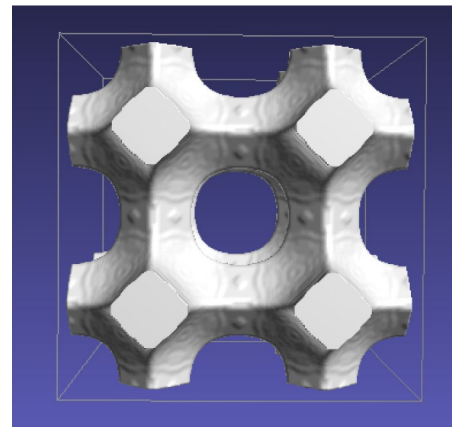
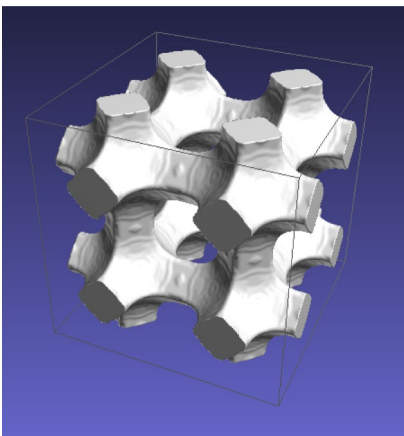


Fig. 9 Training loss of Autoencoder model

(a) One unit cell



(b) 2 by 2 by 2 unit cells

**Fig. 10** Optimized design for linear elastic material model

axial stiffness along the three primary directions, which can be expressed as follows:

$$obj = \text{maximize} \left\{ \frac{1}{3} \cdot (C_{11}^H + C_{22}^H + C_{33}^H) \right\}. \quad (24)$$

In this case, the volume fraction is set at 0.4 for the solid material, and the optimized shape is demonstrated in Fig. 10. The optimization converged after 650 Finite Element Method (FEM) evaluations, and the final optimized objective (*obj*) value reached 0.629.

4.2.2 Metamaterial design for hyperelastic material under large deformations

In this study, we investigate the design of metamaterials made of hyperelastic materials under large deformation, using a simulation method based on the Fast Fourier Transform (FFT) (De Geus et al. 2017). This method is tailored to analyze the mechanical behavior of materials with hyperelastic constitutive laws under periodic boundary

conditions. The simulation is conducted in a three-dimensional (3D) domain, with the representative volume element (RVE) occupying a cubic space and discretized into a regular $40 \times 40 \times 40$ mesh. The material's density field is binarized to represent two distinct material phases. Constitutive laws for Neo-Hookean and Saint–Venant material models are established, where the Neo-Hookean model describes the solid material and the Saint–Venant model represents the void material. The solid material is defined with Poisson ratio and Young's modulus of 0.33 and 69, respectively, and for the void material, the Young's modulus is set to 1×10^{-3} . The Newton–Raphson iterative solver and the Conjugate Gradient (CG) solver are employed to update the deformation state. The overall deformation gradient of the unit cell, denoted as F , is set to

$$F = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \quad (25)$$

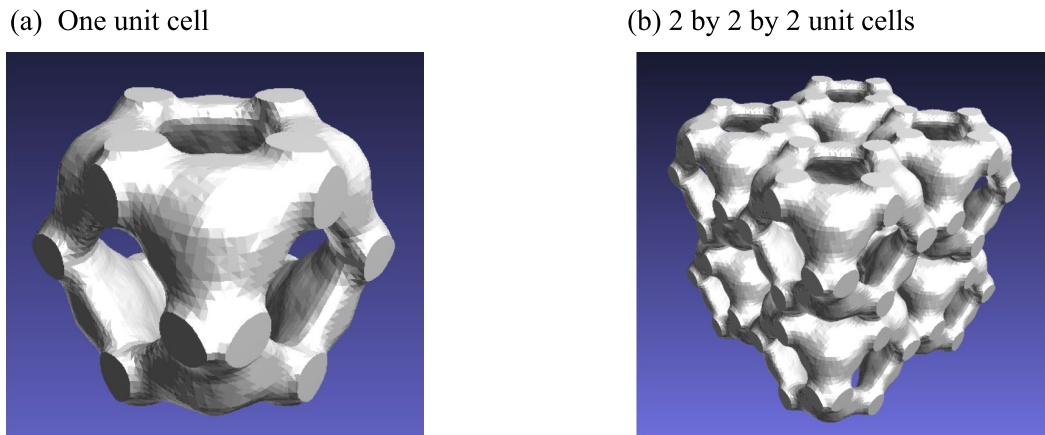


Fig. 11 Optimized design for large deformations with volume fraction 0.35

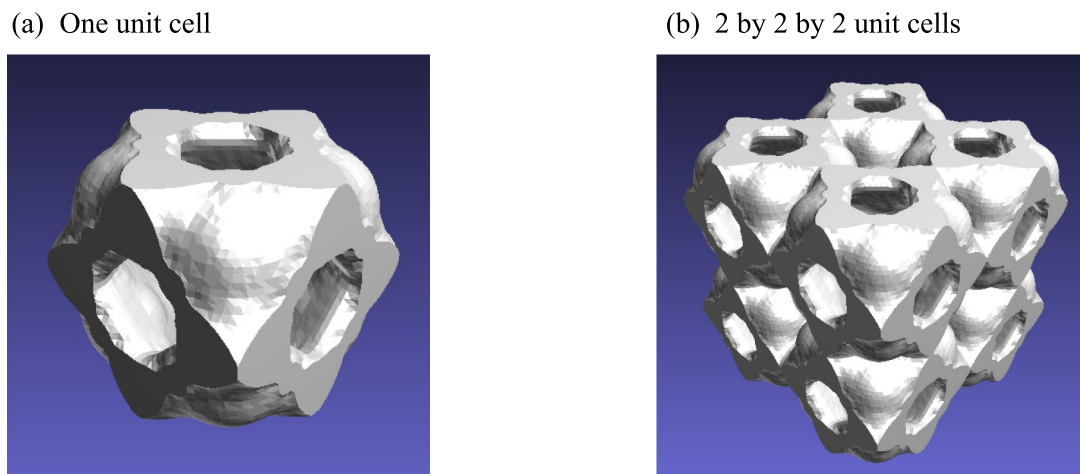


Fig. 12 Optimized design for large deformations with volume fraction 0.5

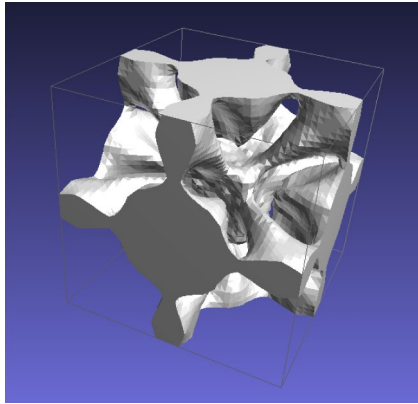
This setting indicates that a 100% strain is applied in each primary direction. The objective function aims to maximize the mean first Piola–Kirchhoff stress along the three primary directions (x, y, z), which can be considered as maximizing the average axial stiffness of the homogenized elasticity tensor. This objective can be written as follows:

$$obj = \max \left\{ \frac{1}{3} \cdot (P_{1111}^m + P_{2222}^m + P_{3333}^m) \right\}. \quad (26)$$

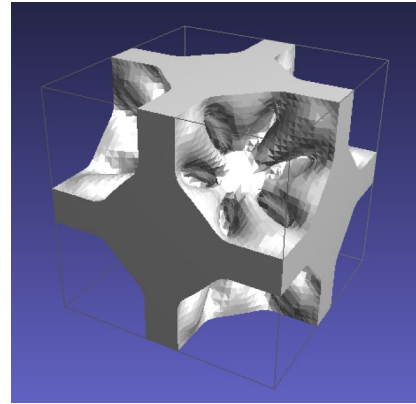
Here, two different volume fractions are applied. The optimized designs with volume fractions of 0.35 and 0.5 are illustrated in Fig. 11. The optimized objective function reached 23.52 for a volume fraction of 0.35, and 29.20 for a volume fraction of 0.5 (Fig. 12). Evidently, the design for large deformation differs from that for linear elasticity, where only small deformations are considered.

For further comparison, we generated two optimized designs under the deformation gradient $F = [3, 0, 0, 0, 3, 0, 0, 0, 3]$, while keeping the material properties unchanged. The optimized designs for two different volume fractions are depicted in Fig. 13. The objective function reached 54.68 for a volume fraction of 0.4 and 50.22 for a volume fraction of 0.3. In Fig. 14, the deformation with periodic boundary conditions for the two different volume fractions is illustrated, where, for ease of visualization, the deformed nodes of the original mesh are displayed. Green points indicate the deformed positions of nodes in solid material, and gray points represent void material. The mechanical responses of the two designs are also presented in Fig. 15, where the x-axis denotes the deformation gradient along the primary direction, and the y-axis represents the mean first Piola–Kirchhoff (PK) stress of the unit cell along the primary direction. Note that the convergence criteria for this case are set to a maximum of 500 evaluations.

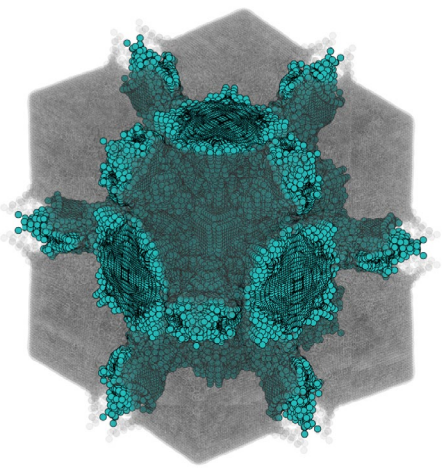
(a) Volume fraction:0.3



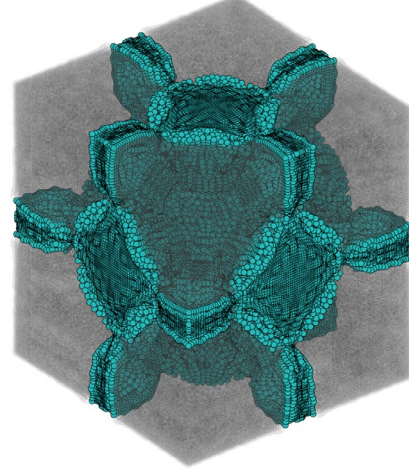
(b) Volume fraction:0.4

**Fig. 13** Optimized design for large deformations

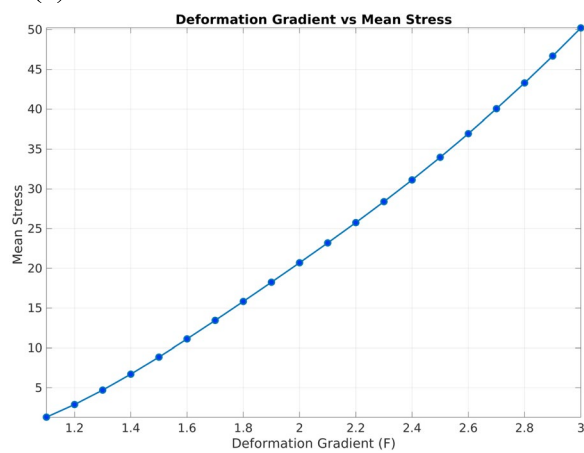
(a) Volume fraction:0.3



(b) Volume fraction:0.4

**Fig. 14** Deformed shape under deformation gradient F **a** Volume fraction:0.3 **b** Volume fraction:0.4

(a) Volume fraction:0.3



(b) Volume fraction:0.4

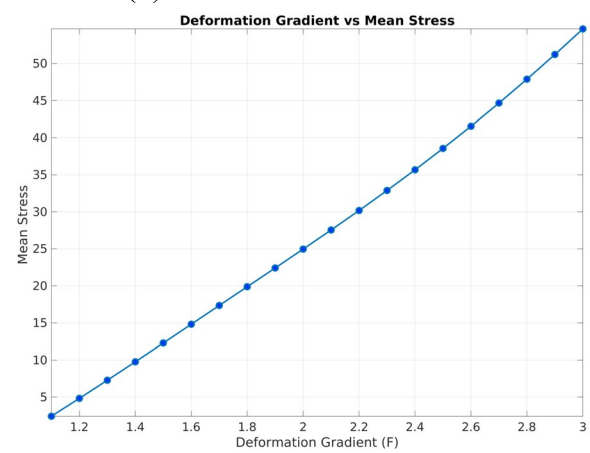
**Fig. 15** deformation response **a** Volume fraction:0.3 **b** Volume fraction:0.4

Figure 15 shows the Piola–Kirchhoff (PK) stress versus Green–Lagrange strain curves for the two optimized designs under uniaxial loading. Both curves exhibit an initially steep slope, reflecting high elastic stiffness, followed by a plateau indicating plastic yielding. The design with higher volume fraction demonstrates greater peak stress and post-yield load retention, as expected. These results confirm that the proposed method can generate designs that effectively leverage material hardening to maintain load-bearing capacity even after yielding. This behavior is critical in real-world applications where structural resilience under nonlinear deformation is required.

4.2.3 Metamaterial design based on elasto-plasticity under finite deformations

In this study, we explore metamaterial design within the context of elasto-plasticity, specifically employing the bilinear model for materials exhibiting two distinct phases with varying properties. This model is solved using an implicit time-integration approach, leading to an elastic-predictor plastic-corrector algorithm within a finite strain framework. Our focus is on a periodic cell, representing a microstructural volume in three dimensions, characterized by the two-phase, elasto-plasticity model. The computational process, similar to the Finite Element Method (FEM), involves a weak form definition, discretization via Galerkin's method, numerical quadrature, and linearization toward a Newton scheme. Notably, the solver's efficiency is attributed to the Fast Fourier Transform (FFT), proving advantageous in handling large deformation mechanical equilibrium problems in periodic unit cells. For further insights into the solver and its mathematical formulations, we refer readers to De Geus

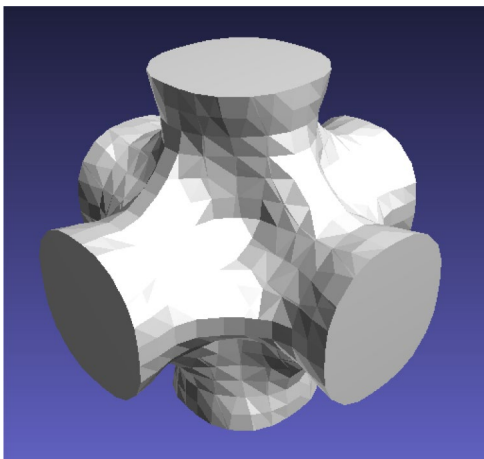
et al. (2017). Here, for computational efficiency, the cubic unit cell is discretized to 30 by 30 by 30 finite elements.

In our material model, we define two distinct materials, A and B, each with unique modulus and yield stress characteristics. The bulk moduli of materials A and B are 0.833 and 1, respectively. The shear moduli for materials A and B are set at 0.386 and 0.42, respectively. The hardening moduli for both materials are set as 0.004 and 0.008, while the initial yield stresses are 0.003 for material A and 0.006 for material B. All parameters are dimensionless. For simplicity, the optimization goal is to maximize the mean consistent tangent stiffness tensor C_{ep} along the primary axes under finite uniaxial tensile strain, with the deformation gradient set at 10% along the x-direction. Loading can only be applied in the x-direction due to assumed geometric symmetry. The objective function can be expressed as follows:

$$Obj = \text{maximize} \left\{ \frac{1}{3} \cdot (C_{ep}(1, 1, 1, 1) + C_{ep}(2, 2, 2, 2) + C_{ep}(3, 3, 3, 3)) \right\}. \quad (27)$$

This formula represents the average of the diagonal components of the C_{ep} tensor, which correspond to the average stiffness along the principal material axes. Note that Material A's volume fraction is maintained at 30%. The optimization, conducted using the BADS method, successfully converged after 400 Finite Element Method (FEM) evaluations. The final optimized objective function, achieved a value of 0.538. The optimized geometry shape of material A is presented in Fig. 16. Additionally, the relationship between the first Piola–Kirchhoff (PK) stress and the Green–Lagrange Strain (E) in the uniaxial tension direction is plotted in Fig. 17. The curve is characterized by a steep initial slope, indicating elastic behavior as the material deforms under increasing

(a) One unit cell



(b) 2 by 2 by 2 unit cells

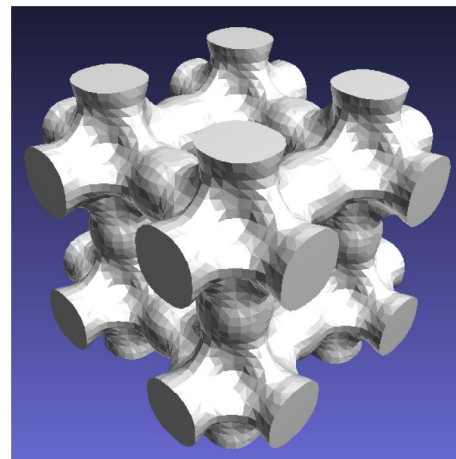
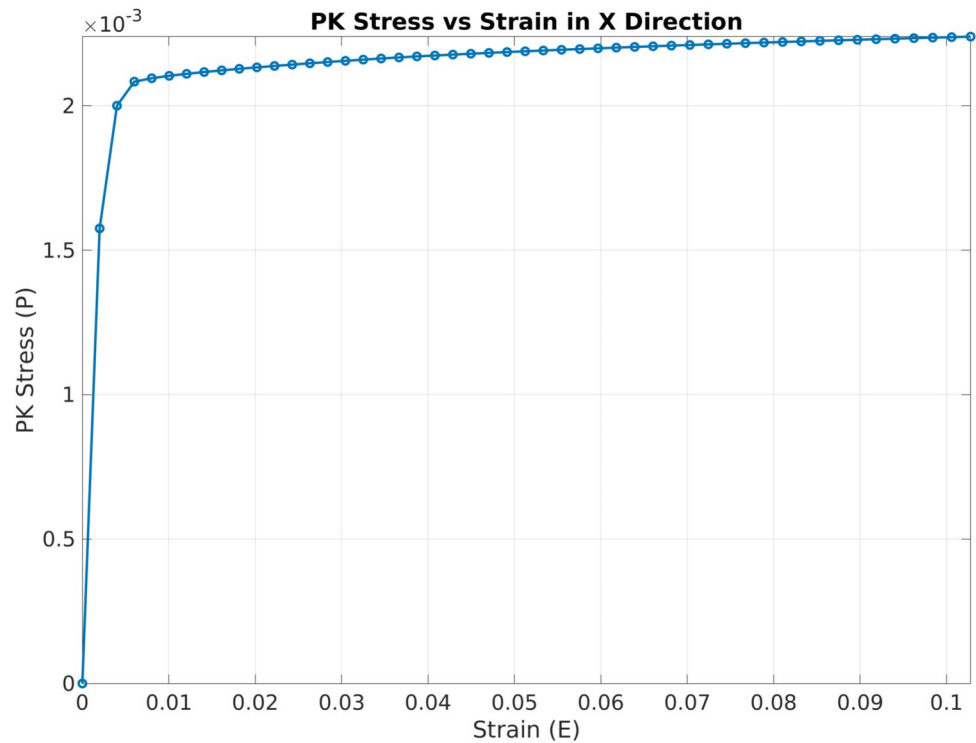


Fig. 16 Optimized design for elasto-plasticity material model

Fig. 17 PK stress vs. Strain in x direction



stress. This elastic region is followed by a plateau, suggesting that the material has yielded and is deforming plastically. The strain values range from 0 to a little over 0.10, and the corresponding first Piola–Kirchhoff (PK) stress values start from 0 and level off just above 0.0020, maintaining this stress level as the strain continues to increase.

It is important to note that for nonlinear problems such as large deformation or elasto-plasticity, there is no unique or “standard” solution due to the inherently non-convex optimization landscape. Different methods may converge to different local optima, depending on initialization and implementation. Our method is not intended to outperform gradient-based approaches, but instead offers a practical and accessible framework, particularly useful in industrial settings where commercial solvers or multi-physics codes are widely used, but gradient information is not available or difficult to compute. By combining a structured design representation with a gradient-free optimization strategy, our goal is to provide engineers and researchers with a feasible tool to solve complex topology optimization problems without relying on adjoint sensitivity analysis or differentiable physics solvers.

In terms of computational cost, the optimization typically converged in 200–300 iterations across the presented cases. Although this is slightly higher than the iteration count of gradient-based SIMP methods (e.g., 100–200 iterations), the difference is largely offset by the fact that

our approach requires no gradient information and is compatible with black-box solvers. Additionally, the dominant computational cost arises from the physics solver per evaluation, not from the optimizer. Our approach offers a more streamlined and solver-agnostic alternative with comparable convergence performance.

5 Conclusion

In this paper, we have proposed an autoencoder method to represent the density field of a unit cell, effectively compressing the initially large design space, where each element functions as a design variable, into a smaller latent space. As demonstrated in both 2D and 3D examples, the autoencoder successfully represents the covariance matrix and compresses it into this compact latent space. With the autoencoder representation, we employ an active learning method to explore the design space. Specifically, the BADS method is applied for active exploration. Initially, we present a 2D unit cell case aimed at maximizing the shear stiffness of the unit cell, where the optimized results prove competitive with standard density field methods, exhibiting similar topology. Furthermore, we extend our investigation to 3D unit cell design, examining three distinct cases: a linear elasticity problem, a large deformation design with hyperelastic material properties, and an

elastic-plasticity problem with finite deformation. Compared to the SIMP method, one of the main advantages of the proposed method is that it is gradient-free. While obtaining gradients for simple problems is straightforward, it becomes significantly more challenging for complex mechanics problems, particularly in nonlinear mechanics, where accurately deriving gradients requires considerable effort. Additionally, most existing FEM solvers, such as MOOSE (Permann et al. 2020) and OpenFOAM (Jasak et al. 2007), do not provide users with arbitrary gradients with respect to design variables, despite having quite comprehensive FEM capabilities for various physical problems. This limitation also applies to commercial software like ABAQUS (2011). Therefore, leveraging these existing solvers for design purposes is where this algorithm proves effective. Overall, these cases all successfully converge and generate effective designs. The results demonstrate that our proposed algorithm can effectively generate meta-material designs for a variety of problems, showcasing its potential to be extended to different physical problems in metamaterial design.

As a part of future work, we aim to quantitatively compare the proposed framework with established nonlinear topology optimization methods such as SIMP with MMA and level-set approaches. While those methods are effective when gradient information is available, our focus is on evaluating how a gradient-free, solver-agnostic strategy performs across a wider range of nonlinear design problems.

Author contributions HD conceptualized the method, implemented the framework, and conducted the numerical experiments. NP and MCM provided methodological guidance and contributed to the manuscript revisions. All authors read and approved the final manuscript.

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Data availability All relevant data, model architecture details, and implementation settings are provided in the manuscript. The authors are willing to share the Python code used for the autoencoder training and optimization upon reasonable request. FEM solver implementations are subject to licensing restrictions and cannot be publicly released.

Code availability The code for the autoencoder and BADS-based optimization framework will be made available upon request.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

Replication of results The numerical examples and training procedures described in this work are based on customized Python implementations. All relevant data, model architecture details, and optimization settings are provided in the manuscript. The authors are willing to share the Python code used for the autoencoder training and BADS-based

optimization upon reasonable request. Due to licensing restrictions, the finite element solver (used for density field evaluation and homogenization) cannot be made publicly available. However, instructions for implementing similar procedures using open-source alternatives such as MOOSE or FEniCS are discussed in the manuscript.

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